Infinite Series 7

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February 2026

Problem 4

Part (a)

Assume for contradiction that the sequence is p-periodic for the smallest possible p. If p is even, then

$$t_{2n} = t_n, \quad t_{2n} = t_{2n+p} = t_{n+p/2},$$

$$t_n = t_{n+p/2}.$$

$$t_{2n+1} = 1 - t_n, \quad t_{2n+1+p} = 1 - t_{n+p/2},$$

$$1 - t_n = 1 - t_{n+p/2} \implies t_n = t_{n+p/2}.$$

In both cases, (t_n) is $p/2$ periodic, which is a contradiction.

If p is odd, let $q = p - 1$. We have that

$$t_{2n} = t_n, \quad t_{2n+p} = t_{2n+q+1} = t_{n+q/2},$$

$$t_n = t_{n+q/2}.$$

$$t_{2n+1} = t_n, \quad t_{2n+1} = t_{2n+1+p} = t_{2n+q} = t_{n+q/2},$$

$$t_n = t_{n+q/2}.$$

So (t_n) is $q/2$ periodic, which is a contradiction. Therefore the sequence is not periodic.

Part (b)

Assume for contradiction that, starting at position k, there is a subsequence of length n where $t_{k+i} = t_{k+i+n} = t_{k+i+2n}$ for all $i \in \{0, ..., n-1\}$. Consider the following cases:

1. If $k + i$ is even:
   (a) If n is even:
   $$t_{\frac{k+i}{2}} = t_{\frac{k+i+n}{2}} = t_{\frac{k+i+2n}{2}}.$$

(b) If n is odd:

$$t_{\frac{k+i}{2}} = t_{\frac{k+i+2n}{2}} = 1 - t_{\frac{k+i+n-1}{2}} = t_{\frac{k+n+i}{2}}.$$

2. If $k+i$ is odd:

   (a) If n is even:

$$t_{\frac{k+i}{2}} = 1 - t_{\frac{k+i-1}{2}} = 1 - t_{\frac{k+i+n-1}{2}} = 1 - t_{\frac{k+i+2n-1}{2}},$$

$$t_{\frac{k+i-1}{2}} = t_{\frac{k+i+n-1}{2}} = t_{\frac{k+i+2n-1}{2}}.$$

   (b) If n is odd:

$$t_{k+i} = 1 - t_{\frac{k+i-1}{2}}, \quad t_{k+i+n} = t_{\frac{k+i+n}{2}} = 1 - t_{\frac{k+i+n-1}{2}}, \quad t_{k+i+2n} = 1 - t_{\frac{k+i+2n-1}{2}},$$

$$t_{\frac{k+i-1}{2}} = t_{\frac{k+i+n-1}{2}} = t_{\frac{k+i+2n-1}{2}}.$$

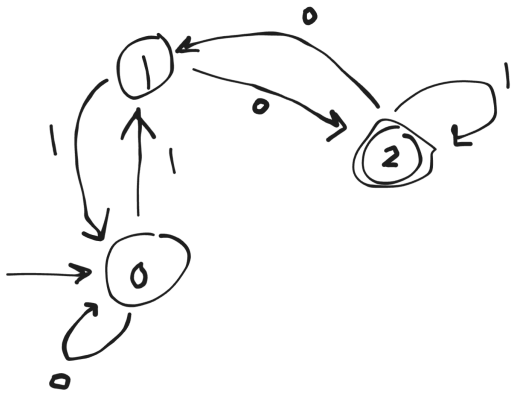
So in every case, we have that there exists the same nonempty subsequence of length $n/2$ or $(n-1)/2$ repeated three times in a row, which is a contradiction.

Problem 10

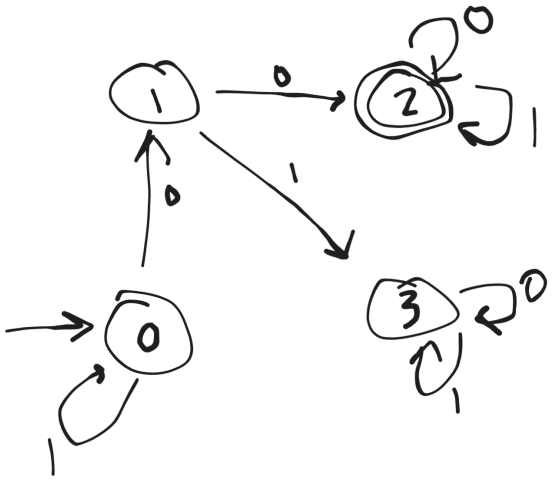
Part (a)

The DFA flips the state if the input is 1 and stays if it's 0. To reach the accept state, the state would need to be flipped an odd number of times. By the definition of the Thue-Morse sequence, this happens exactly when $t_n = 1$.

Part (b)



Part (c)



Problem 11

Part (a)

Figure 1 from the book accepts binary numbers n with an odd number of 1s and rejects otherwise, so the Thue-Morse sequence is 2-automatic.

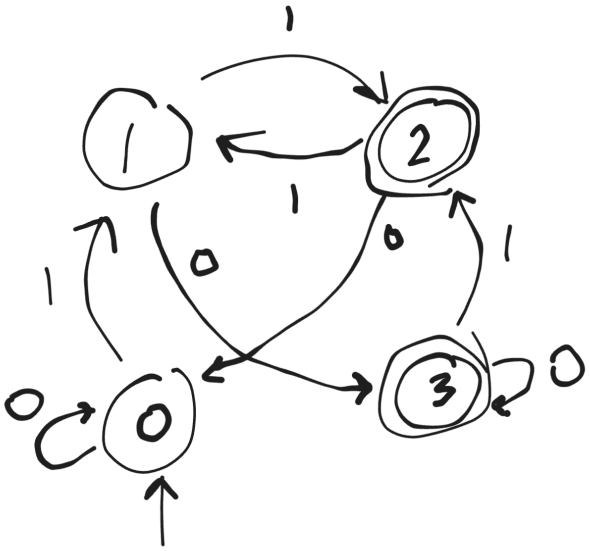
Part (b)

Given the diagram from problem 10c along with the function $f(q_0) = f(q_1) = f(q_3) = 0$, $f(q_2) = 1$, we have that $f(\text{final state}) = 1$ iff $\varepsilon_n = 1$, so the paper-folding sequence is 2-automatic.

Part (c)

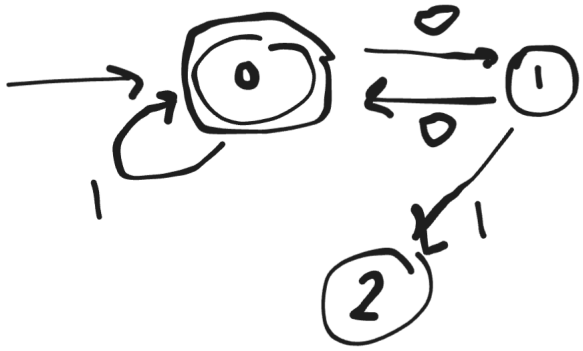
Define a DFA where both states q_0, q_1 are mapped to q_0 when the input is 1 and states are switched when the input is 0. Since the state is swapped whenever a factor of 2 is introduced and resets when a 1 is added, the DFA keeps track of the parity of the number of factors of 2.

Part (d)



The above DFA diagram shows a DFA that accepts n iff the nth term in the Rudin-Shapiro sequence is 1, so it is 2-automatic.

Part (e)



The above DFA diagram shows a DFA that accepts n iff the nth term in the Baum-Sweet sequence is 1, so it is 2-automatic.

Problem 13

Assume a DFA reads base-k digits and outputs a_n. Each base-k digit corresponds to a word of length m in base k, so for each symbol $d \in \{0, \ldots, k^m - 1\}$, we can write d in base k as an m-digit string and define the new transition function to simulate the DFA reading those digits in a sequence.

Now, assume a DFA reads base-k^m digits. To read base-k, we need to group digits into blocks of size m. So the automaton must remember it's current state in the original machine, how many digits of the current m-block it has read, and the value of the block so far. Since m is fixed, that only requires a finite number of states, so the result is still a DFA.